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Neurophysiological signatures of Stanford Neuromodulation Therapy in treatment resistant depression

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Treatment-resistant depression (TRD) affects approximately 30% of patients with major depressive disorder. Stanford Neuromodulation Therapy (SNT), a high-dose intermittent theta-burst transcranial magnetic stimulation protocol, produces rapid antidepressant effects, but its neurophysiological mechanisms remain unclear. Here, we used longitudinal TMS-EEG to characterize the progressive neurophysiological changes induced by SNT, assess their site-specificity, and explore whether baseline neural markers are associated with clinical response. We conducted a double-blind, randomized, sham-controlled trial at Stanford University (2017–2018; analysis August 2024–October 2025) in 24 TMS-naïve participants with TRD (Montgomery-Åsberg Depression Rating Scale ≥ 20 ; ≥ 1 failed antidepressant trial). Participants were randomized to active ($n = 12$) or sham ($n = 12$) SNT, consisting of 10 sessions per day over 5 consecutive days targeting the left dorsolateral prefrontal cortex (90,000 pulses). TMS-EEG was acquired at two baseline sessions, before and after each treatment session, and at 1-month follow-up (14 TMS-EEG sessions in total). Active SNT progressively reduced cortical excitability at the treatment site, with significant decreases by day 3 in the early window component (-27.9% ; $P < 0.01$), while no changes were observed at the vertex control site. Site-specific comparisons confirmed early window reductions only at the left dorsolateral prefrontal cortex ($t_{22} = -3.82$; $P < 0.001$). SNT also selectively decreased estimated medial prefrontal source activity consistent with the subgenual anterior cingulate cortex (sgACC) across sessions ($F_{13,222} = 4.93$; $P < 0.001$), with effects persisting at 1-month follow-up. In an exploratory analysis in the active group ($n = 12$), higher baseline estimated sgACC source activity was associated with greater clinical improvement ($r = -0.67$; $P = 0.023$); although promising, the latter preliminary finding requires replication in larger, adequately powered samples before predictive utility can be established. These findings indicate that SNT induces progressive, site-specific cortical modulation and selective downstream effects on estimated sgACC source activity. Early cortical excitability changes represent candidate neurophysiological markers of SNT response, while the observed association between baseline sgACC activity and clinical outcome, while preliminary, motivates prospective investigation of subcortical source activity as a potential predictor of treatment response in larger trials. ClinicalTrials.gov Identifier: NCT03068715.

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INTRODUCTION

Major depressive disorder (MDD) affects over 322 million people worldwide and represents the leading cause of disability globally [1, 2]. Approximately 30% of MDD patients develop treatment-resistant depression (TRD), a condition where traditional antidepressant treatments like medication and therapy fail to produce meaningful clinical improvement [1]. For patients with TRD, non-invasive brain stimulation techniques such as repetitive transcranial magnetic stimulation (rTMS) targeting the left dorsolateral prefrontal cortex (L-DLPFC) represent a promising alternative treatment approach [3].

Stanford Neuromodulation Therapy (SNT) is a high-dose, accelerated intermittent theta burst stimulation (iTBS) protocol in which the L-DLPFC stimulation site is individually selected to be maximally functionally anti-correlated with the subgenual anterior cingulate cortex (sgACC). In a randomized sham-controlled trial, SNT achieved a 62% mean reduction in depression severity after five days, with a large effect size (Cohen's $d = 1.71$, 95% CI [0.86, 2.56]) delivered over a much shorter timeframe than conventional rTMS protocols which typically require several weeks of treatment [4]. Despite these impressive clinical outcomes, the neurophysiological mechanisms underlying SNT's therapeutic effects still

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require further investigation. Recent studies have begun to elucidate the neural correlates of SNT's antidepressant effects using resting-state functional MRI (fMRI). Batail et al. (2023) reported modulation of functional connectivity involving the default mode network, amygdala, salience network, and striatum following SNT treatment [5]. Similarly, Gajawelli et al. (2024) observed increased anti-correlation between the L-DLPFC and the default mode network following SNT, with changes sustained at one-month follow-up [6]. However, fMRI measures slower hemodynamic responses and cannot capture the immediate millisecond-scale cortical responses to TMS pulses—the direct neurophysiological perturbations that constitute the primary mechanism of action. TMS-EEG provides complementary mechanistic insights by directly measuring evoked cortical responses with millisecond temporal resolution.

Transcranial magnetic stimulation combined with electroencephalography (TMS-EEG) offers millisecond-level resolution of cortical excitability and connectivity [7–11]. TMS-evoked potentials (TEPs), including the P25 (positive deflection at ~25 ms), N45 (negative deflection at ~45 ms), P60 (positive deflection at ~60 ms), N100 (negative deflection at ~100 ms), and P200 (positive deflection at ~200 ms) components, index excitation/inhibition balance and have shown abnormalities in MDD that may normalize with treatment, though findings across studies remain inconsistent [12–14]. Yet, no study has examined the progressive, day-by-day neurophysiological changes that occur during rapid, high-dose SNT treatment, leaving a critical gap in our understanding of how accelerated approaches achieve their rapid and robust clinical effects.

In this study, we used single-pulse TMS-EEG to track neurophysiological changes during the 5-day SNT protocol in patients with TRD, leveraging prospectively collected neurophysiological data from the original double-blind randomized trial. TMS-EEG was recorded at baseline, before and after each treatment day, and at one-month follow-up ($n = 24$; 12 active, 12 sham).

A prerequisite for using TMS-evoked potentials to track treatment-induced neuroplastic changes is that they provide morphologically stable and individually specific measurements across sessions. While prior work has demonstrated test-retest reliability of TEPs in single-session or two-session designs [15, 16], it remains unknown whether this stability holds across an intensive multi-day protocol such as SNT, where cumulative high-dose stimulation could introduce systematic measurement drift. Establishing this reliability is therefore a necessary first step before interpreting any longitudinal amplitude changes as genuine treatment effects. Accordingly, our primary aim was to characterize the progressive, day-by-day cortical excitability changes induced by SNT, as measured by Global Mean Field Amplitude and early local TEPs, and to determine whether these changes are specific to active treatment and to the stimulation site. Our secondary aims were to (1) assess whether SNT modulates source-localized sgACC activity, given its central role in depression circuitry [17–19], (2) determine whether treatment-induced modulation extends beyond the targeted L-DLPFC–sgACC pathway to broader depression-relevant brain networks, and (3) explore whether baseline sgACC activity predicts clinical response—a question motivated by the clinical need for pre-treatment biomarkers given the substantial inter-individual variability in TMS outcomes.

Based on evidence that iTBS induces long-term potentiation-like plasticity [20–22] and that high-dose protocols may produce cumulative plastic changes [23, 24], combined with emerging evidence that clinical efficacy in depression may be associated with cortical response reduction rather than enhancement at stimulation sites [25], we hypothesized that SNT would induce progressive reductions in cortical reactivity, which we interpret as treatment-induced modulation rather than normalization toward healthy control ranges, given the absence of a healthy comparator

group in the present study, and that these changes would persist at post-treatment follow-up.

MATERIALS AND METHODS

Participants

Twenty-four participants (12 active, 12 sham) were included from a double-blind, randomized, sham-controlled trial of Stanford Neuromodulation Therapy (SNT) for treatment-resistant depression (TRD) (ClinicalTrials.gov NCT03068715) [4]. This study represents a secondary analysis of TMS-EEG neurophysiological data collected prospectively as part of the original trial design. While TMS-EEG acquisition was planned from the outset, the specific analytical approaches reported here were developed independently of the primary clinical outcomes. All neurophysiological analyses were conducted blinded to group assignment, with unblinding occurring only after analyses were finalized. The current sample includes 24 of the 29 participants from the original trial who underwent TMS-EEG recordings and had sufficient artifact-free data for longitudinal analysis. All met Maudsley criteria for TRD and had moderate-to-severe depression, defined by the Montgomery-Åsberg Depression Rating Scale (MADRS) scores ≥ 20 [26]. All were TMS-naïve and had failed at least one adequate antidepressant trial. Exclusion criteria included history of seizures, recent substance use disorder, psychosis, unstable medical illness, active suicidal intent, and contraindications to TMS or MRI. Written informed consent was obtained from all participants. The study was approved by the Stanford University Institutional Review Board. The study followed CONSORT guidelines for randomized controlled trials.

Stanford Neuromodulation Therapy (SNT)

SNT consisted of 5 consecutive days of high-dose intermittent theta burst stimulation (iTBS), with 10 sessions per day spaced by 50-minute intervals. Stimulation targeted the L-DLPFC region showing maximal anticorrelation with the sgACC, identified via resting-state fMRI. Each iTBS session delivered 1800 pulses (triplets at 50 Hz repeated at 5 Hz over 2 seconds, 8-second inter-train interval), totaling 90,000 pulses. Stimulation was applied at 90% of resting motor threshold using a MagPro X100 stimulator (MagVenture) with a Cool-B65 A/P coil. Resting motor threshold was determined using neuronavigated single-pulse TMS over the primary motor cortex with depth correction modeling to account for scalp-to-cortex distance [27]. Sham stimulation followed the identical protocol using a sham-capable bidirectional TMS Cool-B65 A/P coil that delivered identical auditory and tactile sensations to active stimulation while blocking magnetic field transmission through a concealed metal plate. The coil was positioned identically to active treatment—held against the scalp at the targeted location using neuronavigation guidance—with the difference between conditions determined by which side of the bidirectional coil was activated (not visible to participants). Both participants and treatment administrators were blinded to treatment assignment through: (1) randomization codes maintained by non-treatment personnel, (2) neutral coil positioning before code entry, (3) lidocaine application as a default procedure at each session, except when participants declined, (4) auditory masking via white noise, and (5) standardized scripts. Treatment administrators were blinded to intensity parameters, and clinical raters were blinded to all treatment information. Post-treatment assessment confirmed participants could not reliably identify their treatment condition above chance.

TMS-EEG data acquisition and processing

Single-pulse TMS-EEG recordings were acquired using the same MagPro X100 stimulator and Cool-B65 A/P coil, with coil positioning guided by the same neuronavigation system targeting the individualized L-DLPFC coordinate, and placement logged at each session. Single-pulse TMS-EEG recordings were acquired at two pre-treatment baseline sessions conducted on the same day, one or two days prior to the first treatment session (with no treatment administered between sessions). On each of the five treatment days, the EEG cap was applied prior to the fifth iTBS session. A single-pulse TMS-EEG recording was acquired before the fifth iTBS session (pre-iTBS) and after the tenth and final iTBS session (post-iTBS), capturing within-day neurophysiological changes across the second half of the daily treatment block (sessions 5–10; 10,800 pulses). Between-day comparisons reflect the cumulative effects of all daily sessions (18,000 pulses per day, 90,000 total). A final recording was acquired at 1-month follow-up. This yielded 14 TMS-EEG timepoints: 2 baseline sessions, 10

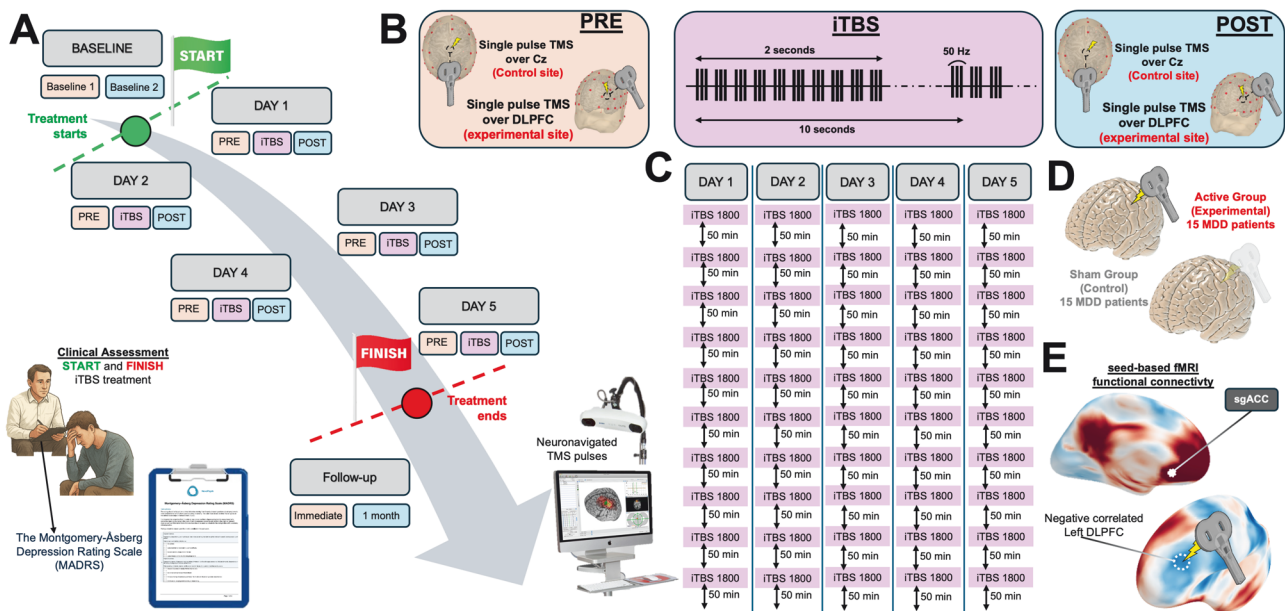


Fig. 1 Stanford Neuromodulation Therapy (SNT) protocol and experimental design. **A** Study timeline illustrating the 5-day accelerated treatment protocol with baseline assessment, daily interventions, and follow-up evaluations. Two pre-treatment baseline TMS-EEG sessions (Baseline 1 and Baseline 2) were conducted with no treatment administered between sessions to establish measurement reliability. The treatment was Stanford Neuromodulation Therapy (SNT). Each treatment day includes pre-intermittent Theta Burst Stimulation (iTBS), and post-iTBS assessment. Pre-iTBS and post-iTBS consisted of single-pulse TMS targeting either the left DLPFC (treatment site) or Cz (control site) for neurophysiological measurements before and after each daily treatment session. Clinical assessments (bottom left) were performed before treatment (baseline), immediately post-treatment, and at 1-month follow-up using the Montgomery-Åsberg Depression Rating Scale (MADRS). **B** Single-pulse TMS procedure targeting either the left DLPFC (treatment site) or the vertex of the scalp (Cz - control site) for comparison. The single-pulse TMS-EEG recordings were acquired every day both pre (orange) and post (light blue) the iTBS protocol (purple). The iTBS protocol consisted of 50 Hz triplet bursts delivered every 200 ms (5 Hz) in 2-second trains followed by 8-second intervals over 10-second cycles. **C** Schematic overview of the SNT protocol which consisted of 10 sessions per day of 1800 pulses per session for a total of 18,000 pulses per day (ISI=intersession interval). **D** Study groups comprising 12 patients with major depressive disorder (MDD) in the active treatment group and 12 patients in the sham control group. **E** TMS targets were individualized using seed-based functional MRI connectivity analysis. The left DLPFC region showing the strongest anti-correlation with the subgenual anterior cingulate cortex (sgACC) was selected as the stimulation target.

treatment-day sessions (pre- and post-iTBS on days 1–5), a post-treatment assessment, and 1-month follow-up (Fig. 1A). The two baseline sessions enabled assessment of test-retest reliability in the absence of treatment. Following EEG cap placement, resting motor threshold was re-measured to account for the increased scalp-coil distance, and single-pulse pulses were delivered at 120% of this adjusted resting motor threshold. EEG was recorded using a 64-channel BrainAmp DC system (Brain Products GmbH), digitized at 5000 Hz, with electrodes referenced to FCz and impedances kept below 5 kΩ. Each session included approximately 150 pulses with jittered inter-trial intervals of 4–6 seconds.

Data preprocessing was performed using the AARATEP pipeline (v2.0), optimized for TMS-EEG artifact removal [28]. Preprocessing included artifact interpolation, ICA-based rejection using ICLabel and TMS-specific criteria, and re-referencing to the average of all channels. Further technical details are provided in the Supplementary Methods.

TEP morphological consistency

Before examining treatment effects, we first needed to establish that TEPs provide morphologically stable measurements across the intensive 5-day protocol, ensuring that any observed amplitude changes reflect genuine treatment-induced modulation rather than measurement drift. TEP morphological consistency was evaluated using cosine similarity [15] measures within and between participants across full TEP epochs (64 channels $\times$ 385 timepoints). Although it is difficult to evaluate TEP reliability during an active treatment protocol that is designed to induce neuroplastic changes, we assessed TEP morphological stability across days during SNT treatment using intraclass correlation coefficients (ICCs). Critically, ICC in this context captures consistency of the topographic spatial pattern and the relative ordering of subjects across sessions rather than absolute amplitude. Because SNT is designed to reduce overall TEP amplitude while preserving the spatial distribution of the evoked response, ICC remains high despite the ~40% amplitude reduction observed in the

active group — the topographic configuration is stable even as the magnitude decreases. This distinction is important: a high ICC here reflects that the same neural generators are being engaged consistently across sessions, not that the response is unchanged. This analysis therefore evaluates whether the overall shape and topographic pattern of TEPs remained consistent across sessions, allowing us to distinguish systematic amplitude changes (reflecting treatment effects) from random variability or morphological drift. ICCs were computed from all measurement sessions and interpreted following established guidelines [29]: values below 0.50 indicate poor reliability, 0.50–0.75 moderate, 0.75–0.90 good, and above 0.90 excellent. Full TEP morphological consistency metrics are available in the Supplementary Methods.

Progressive modulation of cortical excitability

To characterize the overall cortical excitability response to SNT, we evaluated global TEPs. In the present study, global cortical responses were quantified using Global Mean Field Amplitude (GMFA) across three time windows encompassing these components: early (20–80 ms), medium (80–150 ms), and late (150–250 ms) [30–32]. These three components were selected based on established literature demonstrating that they are the most robust and reliably observed TEP peaks in prefrontal cortex stimulation, have been specifically linked to cortical excitability and inhibition mechanisms relevant to depression pathophysiology, and have been previously shown to be modulated by TMS treatment in MDD samples [12, 13, 33].

The area under the curve (AUC) for each GMFA window was computed globally across all 64 channels and normalized to the first pre-treatment baseline session (Baseline 1; Session AUC / Baseline 1 AUC $\times$ 100). Baseline 1 was selected as the normalization reference rather than the average of both baseline sessions because Baseline 2 follows delivery of 150 single TMS pulses during the first probing session; sequential single pulses have been shown to induce cumulative changes in cortical excitability [34],

meaning Baseline 2 may not represent a pure pre-treatment state. Linear mixed-effects models (LME) with TIME (14 timepoints) as a within-subjects factor and GROUP (active vs. sham) as a between-subjects factor were used to test GROUP $\times$ TIME interactions, with random intercepts for subjects to account for within-subject correlations and missing data. Omnibus tests used Bonferroni correction for the three time windows ($\alpha = 0.05/3 = 0.0167$) and post-hoc pairwise comparisons used Bonferroni correction for the fourteen timepoints ($\alpha = 0.05/14 = 0.0036$). Earlier components (< 20 ms) were excluded due to susceptibility to residual TMS-related artifacts and auditory-evoked potentials, while later components (e.g., N300) show high inter-individual variability in latency and amplitude, making them less suitable for group-level longitudinal analysis.

To determine whether treatment effects were spatially restricted to the stimulated cortex or reflected distributed network-level changes, we separately analyzed early local TEPs (EL-TEPs) at the treatment and control sites. EL-TEPs have been shown to be reliable and spatially specific markers of local cortical excitability [10, 11, 35]. The five electrodes closest to the stimulation sites were selected based on Euclidean distance for both the L-DLPFC (treatment site) and the vertex Cz (control site). Early local TEPs were computed as the spatial standard deviation of voltage across these electrode clusters within the [20–80 ms] time window for each stimulation site. Early local TEP analysis was performed separately from global analysis because early components (20–80 ms) reflect spatially restricted, direct cortical excitability at the stimulation site [10, 11, 35], whereas later components represent distributed network responses emerging from recurrent large-scale dynamics [36–39]. This dual approach enabled assessment of both site-specific local effects and broader network-level changes.

To examine topographic changes in TEP amplitude patterns, we performed a two-stage cluster-based permutation testing approach following established procedures [40]. First, within-subject permutation tests (5000 iterations) with cluster-based correction compared post-treatment versus baseline waveforms for each participant separately. For each permutation, trial labels were randomly shuffled, and t -statistics were computed at each channel-timepoint pair. Adjacent channel-timepoint pairs exceeding an initial threshold of $P < 0.05$ (uncorrected) were grouped into spatiotemporal clusters, and cluster-level statistics were computed as the sum of t -values within each cluster. The maximum cluster-level statistic across all clusters was recorded for each permutation to build the null distribution. Observed clusters were considered significant if their cluster-level statistic exceeded the 95th percentile of the permutation-derived null distribution, effectively controlling family-wise error rate at $\alpha = 0.05$. Second, between-group permutation tests (5000 iterations) with identical cluster-based correction procedures compared the magnitude of within-subject changes between active and sham groups across all channels and timepoints. For the between-group analysis, group labels were randomly permuted on each iteration while maintaining within-subject pairing structure. This two-stage approach yielded topographic z -score maps identifying significant spatial-temporal clusters where active SNT produced significantly greater changes than sham treatment, with robust control for multiple comparisons across the entire spatiotemporal data space.

Spatiotemporal dynamics of neurophysiological change

To quantify how consistently each individual participant expressed the group-level spatiotemporal patterns identified above, we applied singular value decomposition (SVD) to extract dominant topographic components from individual data [41]. For each significant cluster identified in the spatiotemporal permutation analysis described above, we extracted 40-ms windows centered on peak latencies (~ 49 ms, ~ 137 ms, ~ 242 ms) from individual participant data. SVD was applied to these windows, and the first spatial component (left singular vector) was extracted as the dominant topographic pattern. We then computed cosine similarity between each participant's dominant spatial pattern and the corresponding group-level difference map. Temporal strength was quantified using the AUC of the first temporal component (right singular vector).

To track the cumulative magnitude of topographic reorganization independently of the direction of change, we computed the Euclidean distance between each session's topographic pattern and the baseline.

sgACC modulation and relationship to clinical response

To examine whether SNT modulates the downstream circuit node most directly implicated in its therapeutic mechanism — the sgACC, which was used to define each participant's individualized stimulation target — we performed source-level analysis of TMS-evoked activity. Specifically, for

sgACC-targeted analysis, source-level activity was extracted from a region of interest defined as Brodmann area 25, inverse-normalized from MNI space to each participant's native brain space, consistent with the sgACC definition used in the original SNT protocol [4]. sgACC responses were quantified as the area under the curve (AUC) across the 20–250 ms post-stimulus window, encompassing both early and late TEP components while excluding periods with high artifact susceptibility. Activation time courses were normalized to the pre-treatment baseline session and analyzed using LME (group $\times$ time) to account for missing data, with Bonferroni-corrected post-hoc testing. Source-localized trials were excluded if the estimated current dipole strength fell below 10% of the maximum observed activation across all timepoints, indicating insufficient signal-to-noise ratio for reliable source estimation.

To determine whether SNT-induced modulation extended beyond the primary L-DLPFC–sgACC circuit to broader functional brain networks, we performed a network-level analysis using the Schaefer atlas [42], which parcellates the cortex into 200 regions organized into seven canonical functional networks: visual (VN), somatomotor (SMN), dorsal attention (DAN), anterior salience (SN), limbic (LN), frontoparietal (FPN), and default mode (DMN). The atlas was projected onto each participant's cortical surface using FreeSurfer spherical registration parameters and resampled to native-space structural T1w MRIs. Source-localized TMS-evoked activity was then averaged across all regions within each network, and treatment-related changes were compared between active and sham groups.

All longitudinal analyses employed linear mixed-effects models implemented in R (version 4.3.1) using the lme4 package version 1.1-35.1 [43]. The model structure was:

$$Y_{ij} = \beta_0 + \beta_1(\text{GROUP}_i) + \beta_2(\text{TIME}_j) + \beta_3(\text{GROUP}_i \times \text{TIME}_j) + u_i + \epsilon_{ij}$$

where Y_{ij} represents the outcome measure for subject i at timepoint j , GROUP is the between-subjects factor (active vs. sham), TIME is the within-subjects factor (14 categorical levels), and u_i represents random intercepts for subjects. Models included random intercepts only, as random slope models did not converge reliably given the sample size. P -values for fixed effects were calculated using Satterthwaite's approximation for degrees of freedom via the lmerTest package version 3.1-3 [44], providing Type III ANOVA F -tests for omnibus effects. Post-hoc pairwise comparisons employed estimated marginal means using emmeans package version 1.8.9 [45] with Bonferroni correction. Model assumptions were verified through residual diagnostics, and all models converged without warnings.

Further technical and statistical details, including preprocessing, denoising thresholds, ROI definition, and SVD parameterization, are reported in the Supplementary Methods.

RESULTS

Spatio-temporal multi-session morphological consistency of TEPs

We first asked whether TMS-evoked potentials provide stable, individual-specific morphological signatures that could reliably track neuroplastic changes across the intensive 5-day SNT protocol. Visual inspection revealed that global TEPs showed consistent morphology across five consecutive treatment days, with stable topographies of canonical components (P25, N45, P60, N100, P200; Fig. 2A). Quantitative cosine similarity analysis confirmed these observations across multiple dimensions. Intraclass correlation coefficients (ICCs) computed across all 14 sessions demonstrated high reliability: $\text{ICC} = 0.81 \pm 0.09$ (mean $\pm$ SD) for the early window component, $\text{ICC} = 0.78 \pm 0.11$ for medium window component, and $\text{ICC} = 0.83 \pm 0.08$ for late window component, all falling within the range of good reliability ($\text{ICC} > 0.75$; [29]). Within-subject cosine similarity maps demonstrated high consistency (yellow regions), while between-subject comparisons revealed lower similarity (blue regions), indicating unique individual neurophysiological signatures (Fig. 2B). Notably, the cosine similarity maps revealed a distinctive ring-shaped region over the stimulated frontal network where within-subject similarity was relatively reduced (upper panels, Fig. 2B) while between-subject similarity in the same region was relatively elevated (bottom panel). This pattern indicates that while TEPs in the stimulated frontal network evolved systematically within each individual across

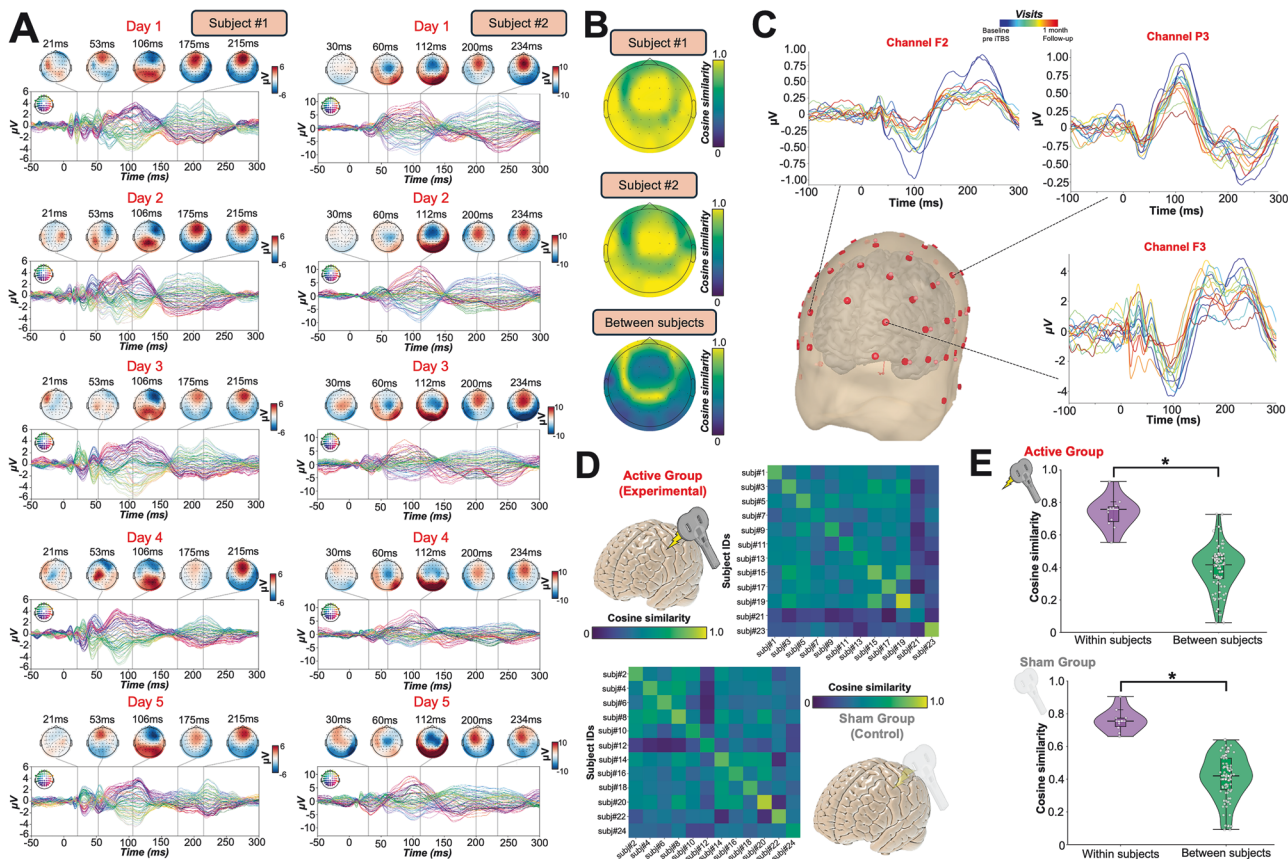


Fig. 2 TEP morphological consistency across multiple visits. **A** TEP topoplots for two representative participants shown across five consecutive pre-treatment days, with topographic maps of TEP components at specific latencies demonstrating consistent spatial distribution of activity patterns across treatment days. This shows that TEP topographies remain stable across multiple sessions, validating the use of TEPs as stable neurophysiological markers. **B** Cosine similarity maps comparing TEP patterns within and between participants. Top panels show within-subject similarity for two representative participants across sessions (high similarity indicated by yellow regions), while the bottom panel shows between-subject similarity (lower similarity indicated by blue regions), demonstrating individual-specific neurophysiological signatures. **C** Morphological consistency of TEP waveforms recorded from three channels: F3 (stimulation target site), P3 (same hemisphere, implicated in the same network), and F2 (contralateral hemisphere) across all sessions for one representative subject, color-coded by visit timing from baseline (blue) to 1-month follow-up (red). TEP morphology is preserved across all recording sites and sessions, confirming that treatment-related amplitude changes reflect genuine neuroplastic modifications. **D** Subject-by-subject similarity matrices for active (top, $n = 12$) and sham (bottom, $n = 12$) groups separately. Diagonal elements represent within-subject similarity across all session pairs, while off-diagonal elements represent between-subject similarity. Both groups show distinct individual TEP signatures with higher diagonal values, confirming that individual specificity is not an artifact of treatment effects. **E** Violin plots quantifying within-subject versus between-subject cosine similarity for active (top) and sham (bottom) groups. Both groups independently show significantly greater within-subject than between-subject similarity (Active: $t = 7.61$, $P < 0.001$; Sham: $t = 7.91$, $P < 0.001$; asterisks indicate $P < 0.001$), demonstrating that TEPs constitute stable, individual-specific neurophysiological markers independent of treatment condition.

treatment sessions (reflected in reduced within-subject similarity as neuroplastic changes accumulated), the spatial topography and direction of these treatment-induced changes were consistent across different participants (reflected in higher between-subject similarity in the same region). In other words, all participants exhibited similar patterns of treatment-related reorganization in the target network, despite maintaining individual-specific baseline signatures in other cortical regions. This observation suggests that SNT induces stereotyped neuroplastic modifications in the targeted frontal-limbic circuitry while preserving individual variability in non-target brain regions. Session-by-session similarity analysis across all 14 sessions revealed excellent consistency (mean cosine similarity = 0.76 ± 0.05), particularly between adjacent days (Fig. 2C). Critically, TEPs demonstrated strong individual specificity: within-subject similarity (0.733 ± 0.104) significantly exceeded between-subject similarity (0.403 ± 0.077 ; $t(20) = 20.81$, $P < 0.001$). When examined separately by treatment group (Fig. 2D–E), both active ($n = 12$) and sham ($n = 12$) groups independently demonstrated significantly higher within-subject than between-subject similarity (Active: within =

0.722 ± 0.127 , between = 0.368 ± 0.134 , $t = 7.61$, $P < 0.001$; Sham: within = 0.752 ± 0.075 , between = 0.380 ± 0.131 , $t = 7.91$, $P < 0.001$), confirming that individual-specific TEP patterns are not artifacts of treatment effects but represent genuine neurophysiological fingerprints that persist across both experimental conditions. This individual specificity suggests that TEPs may potentially serve as reliable neurophysiological markers for tracking systematic changes across treatment protocols, confirming that each participant exhibits a unique, stable neurophysiological signature that is highly consistent within individuals but distinct between individuals [15, 46].

Progressive modulation of TMS-evoked neural responses

We next asked whether SNT produces progressive changes in cortical excitability that distinguish active from sham treatment. We observed differential trajectories of cortical reactivity across treatment sessions in active versus sham groups at the treatment site (Fig. 3A). LME tested GROUP $\times$ TIME interactions for three TEP components at the L-DLPFC treatment site, with Bonferroni correction for multiple comparisons ($\alpha = 0.0167$). The early

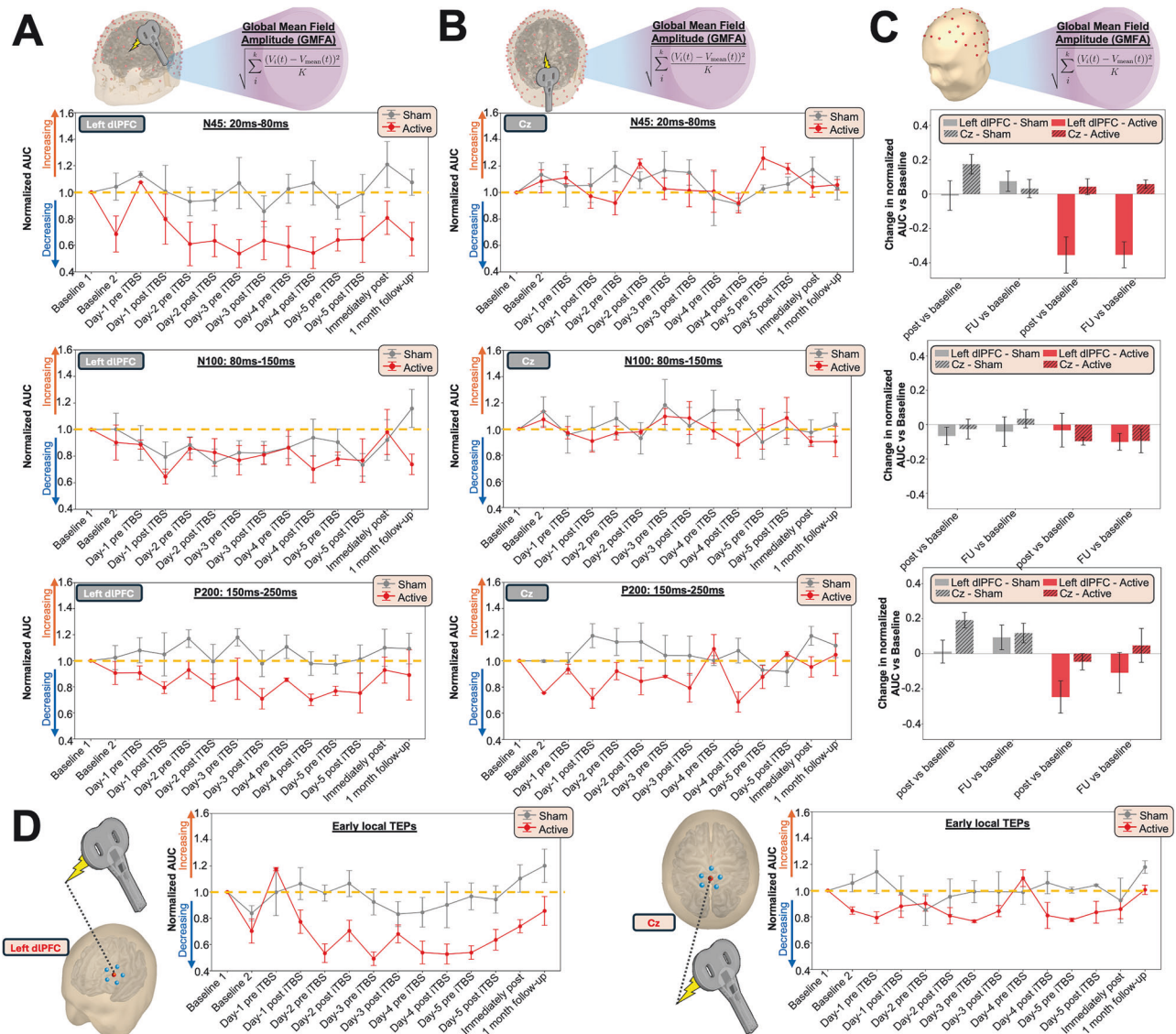


Fig. 3 Progressive modulation of TMS-evoked neural responses following five days of SNT protocol. **A** Normalized area under the curve (AUC) of the global mean field amplitude (GMFA) for three time windows corresponding to established TMS-evoked potential peaks (early window: 20ms-80ms; medium window: 80ms-150ms; late window: 150ms-250ms) recorded from the left DLPFC (treatment site). Data are normalized to individual baseline values. The active SNT group (red) shows a progressive decrease in neural response amplitude for the early window component (20-80 ms) across treatment days. The medium (80-150 ms) and late (150-250 ms) window components did not show significant GROUP $\times$ TIME interactions after Bonferroni correction. The sham group (grey) maintains relatively stable responses across all three time windows. **B** Normalized AUC of GMFA for the same three TEP components recorded from the vertex Cz (control site). Both active (red) and sham (grey) groups show no changes across treatment sessions, demonstrating the site-specificity of the neurophysiological effects observed at the treatment site. **C** Bar plots comparing changes in normalized AUC between treatment (left DLPFC) and control (Cz) sites for both active and sham groups at immediate post-treatment and 1-month follow-up timepoints. The active group shows significant reductions specifically at the treatment site (red bars) for the early window component (20-80 ms), with no significant changes for the medium (80-150 ms) or late (150-250 ms) window components, while showing no significant changes at the control site. The sham group shows no significant changes at either site (grey bars), confirming the treatment-specific and site-specific nature of the neurophysiological modulation. **D** Early local TEPs showing detailed waveforms and time courses for the treatment site (left DLPFC, left panel) and control site (Cz, right panel). The treatment site demonstrates clear reduction of evoked responses in the active group (red) compared to sham (grey), while the control site shows stable responses across both groups, further confirming the spatial selectivity of SNT-induced neuroplasticity.

window component showed a significant interaction (20-80 ms; $F(13,286) = 4.23$, $P < 0.001$, $\eta^2 = 0.10$), surviving correction, as well as main effects of GROUP ($F(1,22) = 8.76$, $P = 0.007$) and TIME ($F(13,286) = 3.92$, $P < 0.001$). In the active group, early window GMFA decreased steadily from baseline (100%) to $72.1 \pm 0.4\%$ by day 3 post-ITBS and $60.2 \pm 0.4\%$ by day 5 post-ITBS, with effects persisting immediately post-treatment ($66.4 \pm 0.5\%$) and partially rebounding at 1-month follow-up ($70.1 \pm 0.5\%$) (Fig. 3A). The sham group showed no systematic change ($98.3\text{--}101.4\%$).

Post-hoc pairwise comparisons (Bonferroni-corrected, $\alpha = 0.0036$) revealed significant between-group differences from day 3 onward. The medium (80–150 ms; $F(13,286) = 1.76$, $P = 0.21$) and late (150–250 ms; $F(13,286) = 1.82$, $P = 0.042$) window components did not show significant GROUP $\times$ TIME interactions after correction, and showed no significant post-hoc differences (Fig. 3A). In contrast, after applying single-pulse TMS at the control site (Cz), no significant changes were observed for any TEP component in either active or sham groups across treatment

sessions (all $P > 0.05$) (Fig. 3B), confirming spatial specificity of treatment effects. This spatial specificity was further confirmed by direct site comparisons showing significant reductions at the treatment site for the early window component ($t(22) = -3.82$, $P < 0.001$) at both immediate post-treatment and 1-month follow-up, while no changes were observed at the control site or for other TEP components (all $P > 0.05$) (Fig. 3C). Early local TEP analysis corroborated these findings, demonstrating significant suppression of evoked responses measured specifically at the treatment site in the active group ($F(13,156) = 3.21$, $P < 0.001$), while local responses at the control site remained stable across both groups ($F(13,156) = 0.83$, $P = 0.65$) (Fig. 3D).

Spatiotemporal dynamics of neurophysiological change and clinical relevance

We characterized the evolution of cortical plasticity during SNT by combining spatiotemporal cluster permutation testing, singular value decomposition (SVD), and trajectory tracking. First, channel-wise cluster permutation tests compared baseline and post-treatment TEPs across all electrodes and timepoints, revealing three significant windows of treatment-related change in the active group (Fig. 4A). The earliest cluster, peaking at ~49 ms (early window component), was centered over frontocentral electrodes (peak $z = -3.42$, $P < 0.01$, cluster-corrected). A second cluster at ~137 ms (medium window component) emerged over central-parietal regions ($z = -3.86$, $P < 0.005$), and a third at ~242 ms (late window component) was widely distributed over frontoparietal

cortex ($z = -4.15$, $P < 0.001$). No significant changes were observed in the sham group. This spatial progression—from localized early effects near the stimulation site to later widespread changes—suggests that SNT may induce plasticity that propagates through connected cortical networks.

To probe individual-level expression of these group-level effects, we applied SVD analysis to assess how closely each subject's neurophysiological changes aligned with the group-defined spatiotemporal signatures of SNT-induced plasticity (see Methods for details).

The temporal amplitude of these subject-specific components was then tracked across the 14-session protocol. In the active group, SVD-derived amplitudes declined to ~60% of baseline by day 5, with partial recovery at 1-month follow-up. The sham group showed no systematic change (Fig. 4B). To quantify treatment-related reorganization and to verify that findings were not an artifact of mixing pre- and post-ITBS sessions within days, we calculated the Euclidean distance between each session's topography and the individual baseline pattern, separately for pre-ITBS and post-ITBS timepoints. Both variants yielded consistent trajectories, confirming the robustness of the observed pattern. Both groups showed an initial increase from baseline to day 1, likely reflecting non-specific factors including practice effects, day-to-day variability, placebo-related changes, regression to the mean, and habituation to the TMS-EEG setup. However, the groups diverged in subsequent days: the active group showed progressive, monotonic increases in Euclidean distance from day 1

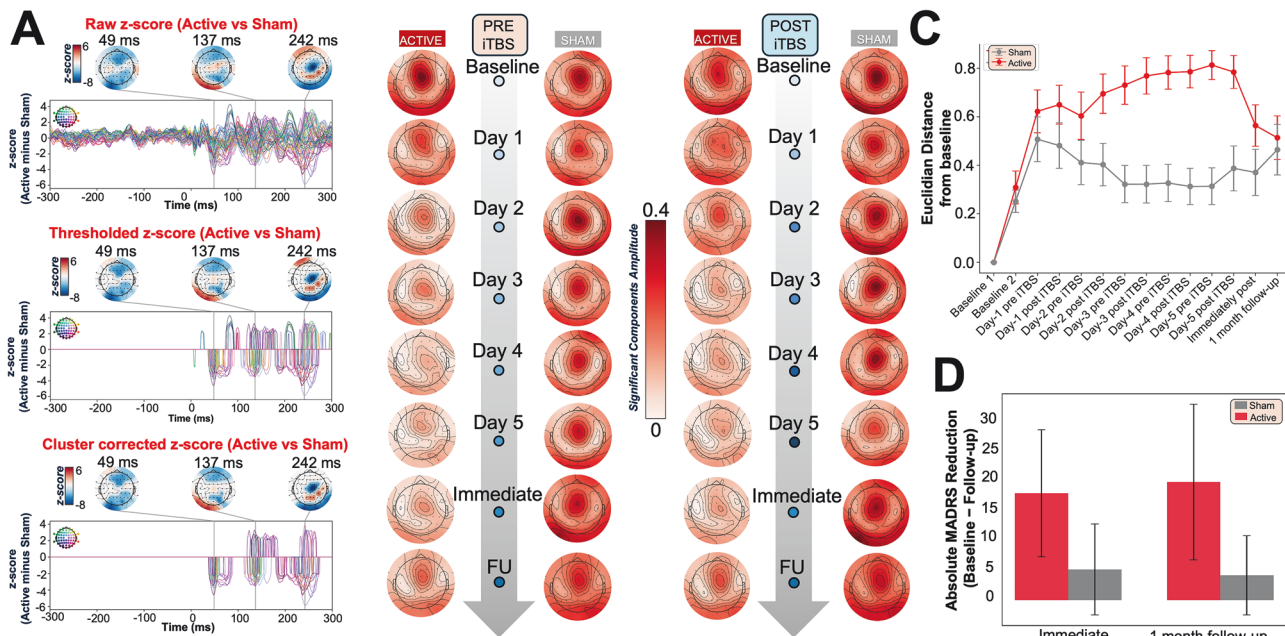


Fig. 4 Topographical neurophysiological changes associated with SNT treatment for TRD. **A** Spatiotemporal cluster-based permutation analysis comparing TMS-evoked responses between first and last treatment visits. Three significant time windows were identified at approximately 49 ms, 137 ms, and 242 ms post-TMS pulse. The analysis shows topographic z-score maps representing statistical differences in neural responses, with negative values (blue) indicating reduced amplitude in the active group compared to sham. The progression from focal frontal effects (49 ms) to more distributed frontoparietal changes (137 ms, 242 ms) suggests intervention-induced plasticity spreading across interconnected networks. **B** Evolution of topographical patterns identified through SVD analysis of the three significant time windows in (panel A), shown separately for pre-ITBS (left panel) and post-ITBS (right panel) sessions. For each condition, active (left column) and sham (right column) topographies are displayed across baseline through day 5 and follow-up. The progressive decrease in neural response amplitude in the active group is consistent across both pre- and post-ITBS sessions, while the sham group maintains stable patterns throughout, confirming that findings are not attributable to mixing of pre- and post-ITBS measurements. **C** Euclidean distance from baseline topographical patterns, quantifying the magnitude of neurophysiological change across treatment days corresponding to the SVD-derived patterns shown in (Panel B). The active group (red) shows progressive increases in distance from baseline, indicating cumulative topographical reorganization, while the sham group (grey) maintains minimal deviation from baseline patterns throughout treatment. Note that both groups show an initial increase from baseline to day 1, likely reflecting non-specific factors; cumulative reorganization in the active group is demonstrated by the progressive increases from day 1 onward, absent in the sham group. **D** Clinical improvements (absolute changes) on the MADRS depression scale were more pronounced in the active group both immediately after treatment and at 1-month follow-up.

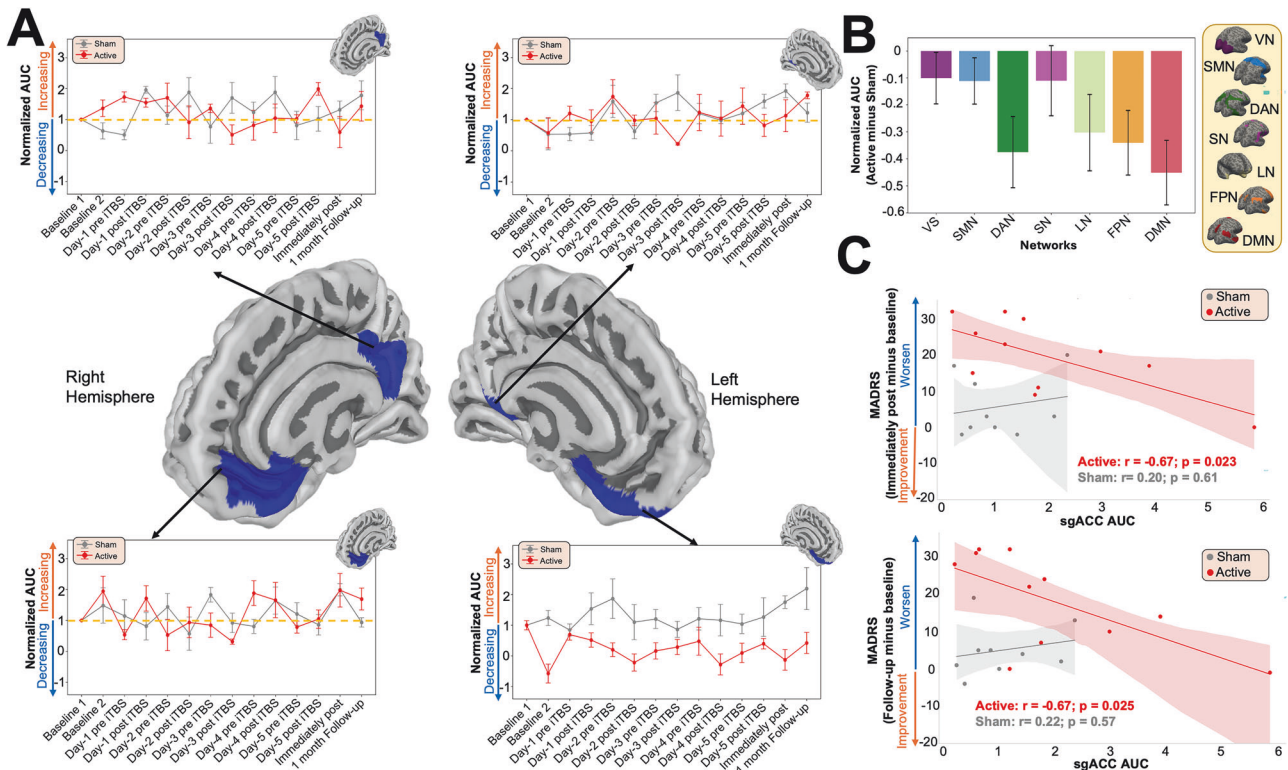


Fig. 5 Regional and network-specific neural responses to SNT and exploratory associations with clinical outcomes. **A** Changes in normalized source-localized area under the curve (AUC) across treatment sessions for the sgACC (the circuit node used to define each participant's L-DLPFC target) and control regions matched for depth from scalp and distance from the therapeutic stimulation site (left DLPFC). Four panels show normalized AUC for these brain areas, with the active SNT group (red) and sham group (grey) displaying distinct trajectories. Significant reduction was observed only in the sgACC following SNT treatment, while control regions show no significant changes, demonstrating the specificity of therapeutic modulation to the intended target circuit. **B** Network-level analysis showing normalized AUC changes for major brain networks including Visual Network (VN), Somatomotor Network (SMN), Dorsal Attention Network (DAN), Salience Network (SN), Limbic Network (LN), Frontoparietal Network (FPN), and Default Mode Network (DMN). Bar plots demonstrate that the active group shows significant reductions in most networks (particularly DAN, LN, FPN, and DMN), while the sham group shows no significant changes across all networks. The widespread network modulation suggests that SNT produces broad-scale neural reorganization beyond the primary sgACC target. **C** Exploratory association between baseline estimated sgACC source activity and clinical outcomes. Scatter plots show the relationship between baseline sgACC AUC and MADRS improvement at immediate post-treatment (top) and 1-month follow-up (bottom). In the active group (red), higher baseline sgACC activity was significantly associated with greater clinical improvement at both timepoints (immediate: $r = -0.67$, 95% CI [-0.90, -0.16], $P = 0.023$; follow-up: $r = -0.67$, 95% CI [-0.90, -0.12], $P = 0.025$), while the sham group (grey) shows no significant relationship. This preliminary finding suggests that higher baseline estimated medial prefrontal source activity consistent with the sgACC may be associated with greater capacity for therapeutic modulation, though this requires replication in larger samples before biomarker utility can be established.

through day 5, whereas the sham group plateaued around the day 1 level without further systematic change (GROUP $\times$ TIME interaction: $F(13,286) = 3.21$, $P < 0.001$; Fig. 4C). This divergent trajectory demonstrates cumulative topographical reorganization specifically in the active treatment group.

We next asked how individual neurophysiological trajectories related to group-level patterns and clinical response. Clinical improvements were significantly greater in the active group compared to sham, both at post-treatment ($\Delta = -7.52$, $t(44) = -4.02$, $P = 0.0002$) and at 1-month follow-up ($\Delta = -6.04$, $t(44) = -3.03$, $P = 0.004$), confirming the sustained antidepressant effect of SNT (Fig. 4D).

Together, these results support a model in which SNT induces progressive and spatially expanding cortical plasticity. Early alignment with treatment-induced topographies may reflect a subject's capacity to engage key neural circuits, offering a potential avenue for biomarker-driven response monitoring and individualized protocol refinement.

sgACC response reduction and its relationship to treatment response

Next, we investigated whether SNT modulated sgACC activity, given its central role in depression and established functional

connectivity with the L-DLPFC [18, 47–49]. Source-localized analysis of sgACC activity (region of interest defined as Brodmann area 25) revealed significant treatment-specific modulation over the SNT protocol. LME modeling demonstrated a significant GROUP $\times$ TIME interaction ($F(13,222) = 4.93$, $P < 0.001$), indicating divergent neurophysiological trajectories between active and sham conditions (Fig. 5A). The active group exhibited immediate and sustained sgACC response reduction beginning from the first post-iTBS session, with activity consistently below normalized baseline levels, while the sham group maintained stable activation throughout treatment.

Bonferroni-corrected post-hoc comparisons confirmed significantly reduced estimated medial prefrontal source activity consistent with the sgACC in the active group at multiple critical timepoints: day 1 post-iTBS ($\Delta = -1.02$, $P < 0.01$), day 3 post-iTBS ($\Delta = -1.65$, $P < 0.001$), immediate post-treatment ($\Delta = -1.88$, $P < 0.001$), and crucially, at 1-month follow-up ($\Delta = -1.77$, $P < 0.001$), demonstrating persistent modulation of this depression-relevant circuit. To control for potential depth-related artifacts inherent to EEG source reconstruction, we analyzed matched control regions at equivalent cortical depths and distance both ipsilaterally and contralaterally to the sgACC.

Control regions were selected at MNI coordinates matched to the sgACC seed in terms of depth from the cortical surface, verified via Euclidean distance from the pial surface and distance from the TMS target, while avoiding regions with known functional connectivity to the DLPFC–sgACC circuit. These control regions showed no significant GROUP $\times$ TIME interactions (all $F < 1.21$, all $P > 0.25$) in either treatment condition, suggesting that the observed sgACC modulation might reflect genuine circuit-specific neuroplasticity.

Having established that SNT selectively modulated sgACC activity, we next asked whether this modulation remained confined to the primary L-DLPFC–sgACC circuit or extended to functionally connected large-scale networks implicated in depression. Network-level analysis revealed widespread but selective modulation across major brain systems (Fig. 5B). The active group demonstrated significant reductions in the Default Mode Network ($\Delta = -0.43$, $P < 0.001$), Frontoparietal Network ($\Delta = -0.38$, $P < 0.01$), Dorsal Attention Network ($\Delta = -0.35$, $P < 0.01$), and Limbic Network ($\Delta = -0.32$, $P < 0.05$), while Visual, Sensorimotor, and Salience networks remained unchanged. The sham group exhibited no significant network-level changes (all $P > 0.05$), indicating that SNT produces coordinated reorganization across depression-relevant circuits beyond the primary sgACC target. However, these network-level comparisons were not corrected for multiple comparisons across the seven networks. Findings should therefore be considered exploratory.

Baseline estimated medial prefrontal source activity consistent with the sgACC showed a preliminary predictive association with treatment response in the active group (Fig. 5C). Higher baseline sgACC activity was associated with greater MADRS improvement at immediate post-treatment ($r = -0.67$, 95% CI $[-0.90, -0.16]$, $P = 0.023$) and at 1-month follow-up ($r = -0.67$, 95% CI $[-0.90, -0.12]$, $P = 0.025$). This relationship was not observed in the sham group (immediate: $r = 0.20$, $P = 0.613$; follow-up: $r = 0.22$, $P = 0.572$), suggesting that baseline sgACC activity may index capacity for therapeutic modulation via the L-DLPFC–sgACC circuit, though replication in larger samples is needed to confirm this specificity.

Active SNT was associated with reduced estimated medial prefrontal source activity consistent with the sgACC across treatment sessions while simultaneously modulating functionally connected networks. Exploratory analysis suggested an association between baseline sgACC activity and treatment response ($r = -0.67$). However, given the modest sample size ($n = 12$), lack of correction for multiple correlational tests, and exploratory nature of this analysis, these findings should be considered preliminary. Replication in adequately powered, independent samples with pre-specified hypotheses is essential before any clinical application.

DISCUSSION

Using comprehensive daily TMS-EEG measurements during SNT in treatment-resistant depression, we identified several key neurophysiological signatures of therapeutic response. First, TMS-evoked potentials demonstrated excellent morphological consistency and individual specificity across the intensive 5-day protocol, validating their use as stable neurophysiological markers. Second, SNT induced progressive modulation in cortical excitability, with early-window GMFA decreasing by 27.9% by day 3 and reaching ~40% response reduction by treatment end. Third, spatiotemporal cluster analysis revealed three distinct windows of treatment-related change (49 ms, 137 ms, 242 ms), with effects progressing from localized frontal regions to distributed frontoparietal networks. Finally, source-localized analysis demonstrated selective and sustained reduction of estimated medial prefrontal source activity consistent with the sgACC throughout active treatment, with baseline estimated source activity showing a

preliminary association with clinical response ($r = -0.67$). A critical advance of this study is the demonstration that therapeutic neuroplastic changes in accelerated TMS emerge through progressive, cumulative mechanisms rather than abrupt threshold effects, with systematic day-by-day evolution that provides temporal precision for understanding plasticity kinetics and optimizing treatment protocols. Together, this work suggests that SNT's rapid antidepressant effects are associated with cumulative neurophysiological changes in depression-relevant circuits serving as promising biomarkers for personalizing TMS treatment. Importantly, the observed modulation in cortical excitability may reflect normalization of pathologically elevated activity rather than suppression below physiological levels. Patients with TRD exhibit baseline hyperexcitability in frontal-limbic circuits [13], and our findings are consistent with the possibility that SNT progressively shifts dysregulated activity toward normative ranges, potentially through homeostatic plasticity mechanisms, though this interpretation requires direct experimental validation. This normalization framework, rather than simple suppression, may better account for both the therapeutic efficacy and safety profile of iTBS across psychiatric disorders.

Our main finding reveals that progressive modulation in cortical excitability, indexed by the early window TEP component, is associated with SNT's therapeutic effects. Whether this reflects normalization of pathologically elevated E/I balance remains a mechanistic hypothesis consistent with the data rather than a conclusion directly established by them. The active group showed a 27.9% reduction in early-window GMFA by day 3 and ~40% by treatment end, while sham treatment produced no systematic changes (<5%). This temporal evolution aligns with SNT's rapid antidepressant effects [4], indicating cumulative neuroplastic changes that build across treatment sessions rather than immediate effects. The progressive cortical response reduction mirrors the timeline of clinical improvement reported in the original trial, where significant MADRS reductions emerged by day 3–4. This temporal concordance suggests an association between neurophysiological changes and symptom improvement, though the directionality and causal nature of this relationship cannot be established from our observational data. Group-level early-window GMFA reduction reached ~40% by day 5, alongside a 62% group-level mean reduction in depression severity; individual-level correlations between neurophysiological and clinical changes were not tested and require further investigation. The early-window reduction was driven substantially by the N45, a component linked to GABA-A-receptor-mediated inhibitory neurotransmission and cortical excitation/inhibition balance [12]. Prefrontal E/I dysregulation, including altered TMS-EEG inhibitory and excitatory markers, has been reported in MDD [13], and DLPFC TMS-EEG markers have been associated with fronto-limbic circuit function [50]. Interpreted within this framework, the concurrent reduction in early-window/N45 reactivity and in estimated sgACC source activity is consistent with treatment-induced normalization of an over-reactive prefrontal-limbic circuit. Because GMFA integrates overlapping components across the 20–80 ms window, however, a specific attribution to GABAergic inhibition cannot be established here and remains a hypothesis requiring pharmacological and higher-resolution confirmation. The concurrent modulation of the late window component, reflecting network-level propagation [36, 38, 39], suggests SNT affects both local cortical excitability and broader information processing networks. Neurophysiological changes persisted at 1-month follow-up with partial recovery (from 60% to 70% of baseline), indicating long-term depression (LTD)-like plasticity mechanisms. This observation is particularly noteworthy given that iTBS is conventionally characterized as an 'excitatory' protocol based on motor cortex studies. However, our findings—showing sustained reduction rather than enhancement of cortical excitability following therapeutic iTBS—align with accumulating

evidence from motor cortex studies demonstrating that iTBS effects are considerably more complex than simple excitation [51, 52]. This partial recovery corresponds with the maintenance of clinical gains reported in the original trial, consistent with a potential biological basis for sustained clinical efficacy following multi-session therapeutic TMS protocols.

The apparent paradox between single-session ‘excitatory’ iTBS effects in motor cortex and our observed cumulative cortical response reduction in prefrontal cortex during therapeutic SNT may be explained by several non-mutually exclusive mechanisms. First, synaptic metaplasticity—whereby the history of prior synaptic activity determines the direction and magnitude of subsequent plasticity—may play a critical role in multi-session accelerated protocols [53, 54]. In accelerated SNT, the cumulative dose of 90,000 pulses over 5 days may engage homeostatic mechanisms that progressively shift the excitation-inhibition balance toward normalization rather than simple potentiation. Second, recent evidence from both empirical studies [13] and computational modeling [25] suggests that therapeutic effects in depression may specifically arise from restoring dysregulated excitation-inhibition balance through preferential modulation of inhibitory circuits, resulting in net cortical response reduction despite nominally ‘excitatory’ stimulation parameters. Third, the prefrontal cortex may respond fundamentally differently to theta-burst stimulation compared to motor cortex due to distinct cellular composition, receptor distributions, and network architecture. The convergence of progressive cortical response reduction across multiple independent iTBS depression studies [13, 25, 33, 55, 56] is consistent with therapeutic mechanisms involving normalization through inhibition rather than simple excitatory enhancement. Even though this remains an interpretation that requires direct mechanistic investigation, it might highlight the need to move beyond single-session motor cortex models when interpreting prefrontal therapeutic interventions.

Our spatiotemporal analysis reveals that SNT induces spatially structured neurophysiological changes, beginning focally at the stimulation site and propagating progressively to distributed frontoparietal networks, consistent with transsynaptic propagation through interconnected cortical circuits rather than uniform global suppression. Cluster permutation testing identified three distinct windows of treatment-related change at 49 ms, 137 ms, and 242 ms post-stimulation, each with unique topographical distributions. The earliest effects (49 ms) were localized over frontocentral regions near the stimulation site, while later components showed progressively more distributed patterns extending to central-parietal (137 ms) and frontoparietal networks (242 ms). This temporal-spatial progression supports transsynaptic propagation through interconnected cortical networks [37, 38, 57], consistent with contemporary views of TMS as a network-level intervention that selectively engages circuits based on stimulation parameters [58]. SVD analysis further demonstrated that individual participants’ neurophysiological changes aligned with these group-level patterns, with spatial components showing monotonic amplitude reduction to ~60% of baseline by day 5. The progressive nature of these changes, quantified through Euclidean distance metrics, revealed cumulative cortical reorganization that was absent in sham participants, confirming the treatment-specific nature of the observed plasticity. Notably, both groups exhibited an initial increase in Euclidean distance from baseline to day 1, likely reflecting non-specific factors common to early treatment exposure including habituation to the protocol, reduced anxiety, normal cortical variability, and potential placebo effects. The cumulative topographical reorganization is evidenced not by this initial shared change, but by the subsequent progressive divergence: systematic monotonic increases in the active group versus plateau in the sham group across days 2–5. The 50-minute inter-session intervals used in SNT may be important for enabling cumulative plasticity effects, consistent with evidence that spaced

learning paradigms enhance synaptic consolidation [59, 60], though direct comparisons with alternative intervals are needed to confirm this hypothesis.

Finally, our EEG source-reconstruction analysis showed how sgACC might play a key mechanistic role in SNT’s therapeutic effects. Notably, the L-DLPFC stimulation site was specifically selected based on its maximal anti-correlation with the sgACC in each individual participant, making sgACC responses a particularly relevant measure of circuit-specific treatment effects. The active group demonstrated progressive sgACC response reduction beginning from the first treatment session and persisting through 1-month follow-up, while sham participants maintained stable activity throughout. Given the well-established role of sgACC hyperactivity in depression pathophysiology [17, 18, 48, 49, 61], this selective reduction is consistent with therapeutic modulation of depression-relevant circuitry. Critically, baseline estimated medial prefrontal source activity consistent with the sgACC showed a preliminary association with clinical response ($r = -0.67$), independent of baseline symptom severity, suggesting that sgACC hyperactivity may index neural circuit capacity for therapeutic modulation rather than simply reflecting current depression severity. This finding supports contemporary circuit-based models of depression, where functional connectivity to sgACC predicts TMS efficacy across different stimulation targets [18], and sgACC-guided targeting strategies improve clinical outcomes [48]. Our results align with this framework, suggesting that elevated baseline estimated medial prefrontal source activity consistent with the sgACC may be associated with greater potential for therapeutic modulation via the L-DLPFC–sgACC circuit. This association derives from a small single-site active-arm sample ($n = 12$) with uncorrected exploratory testing and should be regarded as hypothesis-generating; it requires prospective replication in larger, adequately powered, independent cohorts before any predictive utility can be claimed. It is important to report that the sgACC source localization was performed using 64-channel EEG, which has limited spatial resolution for deep medial structures. Although the absence of significant effects in depth-matched control regions argues against a generic source reconstruction artifact, the spatial precision of these estimates remains inherently constrained. However, prior concurrent EEG–PET work has shown that LORETA source estimates of medial prefrontal and rostral/subgenual anterior cingulate activity are spatially concordant with FDG-PET measures, even with lower-density montages [62, 63], providing cross-modal support for estimating deep medial source activity from scalp EEG. Future studies using high-density EEG (128–256 channels) or concurrent fMRI would provide stronger spatial confidence for sgACC-specific claims.

Our findings are consistent with a homeostatic metaplasticity framework in which repeated TMS sessions induce cumulative plastic changes that shift pathologically dysregulated circuits toward normative excitation-inhibition balance [53, 64]. Under this theoretical proposal, the direction of plasticity would be determined by the baseline state of the circuit rather than by the nominal excitatory or inhibitory parameters of the stimulation. This remains a speculative interpretation of the present data and requires direct experimental testing across dose conditions, patient populations, and psychiatric indications before any broader conclusions can be drawn.

First, it may provide a mechanistic basis for understanding why TMS demonstrates efficacy across a remarkably diverse range of neuropsychiatric conditions beyond MDD—including obsessive-compulsive disorder, post-traumatic stress disorder, substance use disorders, and anxiety disorders—despite vastly different symptom profiles and presumed underlying pathophysiology. If TMS fundamentally operates through homeostatic normalization rather than unidirectional excitation or inhibition, it could adaptively restore circuit function across different forms of dysregulation. The

key determinant of efficacy may become whether a given circuit exhibits pathological deviation from normative excitability that can be modulated through the targeted network, rather than whether it requires specific ‘excitatory’ or ‘inhibitory’ intervention.

Second, this framework may explain the characteristic bimodal distribution of TMS treatment outcomes: patients tend to be either responders (those whose neural activity normalizes with cumulative sessions, corresponding to clinical improvement) or non-responders (those whose neural activity fails to normalize, corresponding to persistent symptoms), while ‘anti-responders’ showing progressive clinical worsening are exceptionally rare. Under this theoretical framework, homeostatic metaplasticity mechanisms could constrain the direction of change toward physiological ranges, though this remains speculative and requires direct experimental testing across dose conditions and patient populations. Non-response likely reflects insufficient engagement of the pathological circuit (due to anatomical variability, connectivity patterns, or biological constraints on plasticity capacity) rather than maladaptive plasticity in the wrong direction. Our exploratory finding that higher baseline estimated medial prefrontal source activity consistent with the sgACC was associated with greater treatment response is compatible with this interpretation: patients with greater deviation from normal excitability may have more ‘room’ for normalization and may be more likely to engage homeostatic mechanisms.

These observations have important implications for both understanding TMS versatility across indications and its favorable safety profile. The normalization framework suggests that identifying the relevant dysregulated circuit and ensuring adequate engagement through personalized targeting based on functional connectivity-guided localization represents a theoretically motivated strategy for circuit engagement, though whether it confers advantages over standard group-based approaches remains an open empirical question that the present dataset cannot resolve. Future research should explicitly test this normalization hypothesis across different psychiatric conditions and baseline states, potentially enabling more principled approaches to treatment selection and parameter optimization.

Overall, our findings reveal an apparent discrepancy between the nominal physiological effects of iTBS and its therapeutic mechanisms in depression. While iTBS has been classically characterized as ‘excitatory’ based on early motor cortex studies showing increased motor-evoked potentials [20–22], substantial inter-individual variability has been documented, with studies reporting fairly even distributions of facilitatory and inhibitory responses across individuals for both iTBS and cTBS [51]. This variability may reflect the influence of baseline cortical state, receptor-dependent homeostatic metaplasticity mechanisms [53, 54], and differences between acute single-session and cumulative multi-session protocols. This pattern of excitability reduction, rather than enhancement, predicting clinical improvement has been observed across multiple iTBS depression protocols [25, 33], suggesting a fundamental difference between motor and prefrontal cortex responses to theta-burst stimulation. From a circuit perspective, this apparent paradox may reflect preferential modulation of inhibitory interneurons (resulting in net inhibition despite LTP-like mechanisms) or induction of long-term depression-like plasticity in excitatory neurons at the prefrontal target [65]. Further investigation of cell-type-specific and circuit-level mechanisms underlying iTBS effects in prefrontal cortex is needed to reconcile these observations with established motor cortex findings.

These neurophysiological signatures are consistent with a precision medicine framework for TMS, though prospective validation in larger samples and direct comparisons with standard targeting approaches are needed before clinical translation. If prospectively validated, baseline sgACC-related measures could eventually help stratify candidates for SNT. However, the present

study does not establish clinical utility or validate task-free fMRI as an interchangeable measure. Longer-term integration of portable EEG systems could enable routine biomarker-guided TMS monitoring across diverse clinical settings, maximizing therapeutic efficacy while minimizing treatment burden for predicted non-responders. Moreover, future studies should apply similar dense longitudinal TMS-EEG approaches to compare neurophysiological mechanisms across different TMS protocols (e.g., conventional daily rTMS versus accelerated protocols) and targets, to determine whether therapeutic mechanisms are common across approaches or protocol-specific, which could inform optimization of treatment parameters and patient-protocol matching strategies.

Several limitations should be noted. The sample size of $n = 12$ per group is modest relative to the analytic complexity employed, and while the repeated-measures design substantially increases statistical efficiency for within-subject inferences, it does not fully compensate for instability in between-subject comparisons. The correlation between baseline sgACC activity and clinical response ($r = -0.67$) is exploratory and uncorrected, and should be treated as hypothesis-generating rather than confirmatory; thus, replication in larger prospective cohorts is required before this relationship can inform clinical decision-making. The lack of direct L-DLPFC–sgACC functional connectivity measurements further limits mechanistic interpretation of the circuit-level findings. Additionally, source localization of deep cortical and subcortical structures such as the sgACC using scalp EEG remains inherently challenging due to limited spatial resolution and inverse solution constraints. Accordingly, interpretations of sgACC-related activity should be considered indirect estimates rather than direct measurements of regional neural activity. In addition to this, auditory masking was not applied during single-pulse TMS-EEG probing sessions, which may have introduced auditory evoked potential contributions to the N100 and P200 time windows. However, the inclusion of a sham control arm and a vertex control site argues against a generalized auditory artifact explanation, as such contributions would be expected to affect both conditions and both sites equivalently. Findings pertaining to later components should nonetheless be interpreted with this caveat in mind.

Lidocaine was applied as a default procedure at each session to reduce scalp discomfort and preserve sham blinding, with exceptions only when participants declined. While application was not formally logged per session per participant, the near-systematic nature of its use limits — though does not fully exclude — the possibility that session-to-session variability in somatosensory afference contributed to TEP amplitude changes. Future studies should formally document anesthetic application across sessions.

The study lacked comparison conditions using either standard group-based DLPFC targeting or alternative TMS protocols. It is therefore not possible to determine whether the observed neurophysiological effects are specific to SNT — including its accelerated schedule, high pulse dose, and individualized connectivity-guided targeting — or reflect generic effects of repeated high-dose prefrontal iTBS, and their generalizability to conventional TMS remains uncertain. Direct comparative studies are needed to resolve this. These findings nonetheless provide a foundation for biomarker-driven TMS protocols that could improve treatment outcomes in TRD.

Mechanistic interpretations beyond the single corrected component — including multi-component normalization and E/I balance restoration — remain exploratory hypotheses consistent with the data rather than established conclusions, and should be treated with caution pending replication in larger samples.

Finally, the absence of a healthy control group limits our ability to determine whether the observed reductions in cortical excitability represent a shift toward healthy ranges or suppression below normal physiological levels. Future studies should include healthy control groups measured longitudinally with identical TMS-EEG protocols to establish normative ranges.

In conclusion, SNT is associated with progressive modulation in cortical excitability and estimated medial prefrontal source activity consistent with the sgACC across the treatment course, neurophysiological changes consistent with — though not directly establishing — a mechanism underlying its therapeutic effects in treatment-resistant depression. Higher baseline estimated medial prefrontal source activity consistent with the sgACC was associated with better clinical outcomes in an exploratory analysis, although these findings require validation. These neurophysiological signatures advance our understanding of accelerated TMS mechanisms and may inform future personalized approaches for treatment-resistant depression, potentially enabling the identification of optimal candidates and treatment parameters to maximize therapeutic efficacy. These findings have not yet been replicated in an independent cohort or across multiple centers, and therefore require external validation before broader generalization can be supported.

DATA AVAILABILITY

The clinical trial data are registered under ClinicalTrials.gov identifier NCT03068715. De-identified TMS-EEG data supporting the findings of this study are available from the corresponding author upon reasonable request. Preprocessing was performed with the publicly available AARATEP pipeline (v2.0). Analysis code is available from the corresponding author upon request.

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AUTHOR CONTRIBUTIONS

NW conceived and designed the study. NJJ, EJC, AM-K, and CT acquired the data. DM performed the analyses. DM wrote the initial draft. DMB, MW, ADG, CCC, DAP, CJK, and CER revised the manuscript and contributed to the revisions. CJK, CER, MS, DV, JDG, and NW supervised the work. All authors contributed to interpretation of the results, critically revised the manuscript, and approved the final version.

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COMPETING INTERESTS

Dr. Williams holds patents for the SNT protocol and has equity in Magnus Medical, which has licensed the SNT technology. Dr. Williams has served on scientific advisory boards for Otsuka, Jazz Pharmaceuticals, and Janssen. Dr. Keller holds equity in Alto Neuroscience, Inc. and Flow Neuroscience, Inc. Over the past three years, Dr. Pizzagalli has received consulting fees from AbbVie, Boehringer Ingelheim, Circular Genomics, Compass Pathways, Engrail Therapeutics, Neumora Therapeutics, Neurocrine Biosciences, N1 Biocorp., Neuroscience Software, Syntropic Medical, Takeda, and Xenon Pharmaceuticals; he has received honoraria from the American Psychological Association, Psychonomic Society and Springer (for editorial work) and from Alkermes; he has received research funding from the Brain and Behavior Research Foundation, Circular Genomics, Dana Foundation, Wellcome Leap, Millennium Pharmaceuticals, and NIMH; he has received stock options from Ceretype Neuromedicine, Compass Pathways, Engrail Therapeutics, Neumora Therapeutics, and Neuroscience Software. No funding from these entities was used to support the current work, and all views expressed are solely those of the authors. No other authors report conflicts of interest.

ETHICS

The study was approved by the Stanford University Institutional Review Board and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants prior to enrollment. The trial was registered at ClinicalTrials.gov (NCT03068715) and followed CONSORT guidelines for randomized controlled trials.

ADDITIONAL INFORMATION

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